

10.7 WEYL ORDERING AND THE MIDPOINT RULE

A classical function $f(q, p)$ does not determine a unique quantum operator: qp could become QP , PQ , or any combination that differs from these by multiples of $[Q, P] = i\hbar$. **Weyl ordering** is the symmetric choice. It is defined on exponentials by

$$e^{xp+kq} \longmapsto e^{xP+kQ},$$

and extended to any f by writing f as a superposition of exponentials (a Fourier integral). For example $qp \mapsto \frac{1}{2}(QP + PQ)$. The function $f(q, p)$ is called the **Weyl symbol** of the operator. Below x and k are parameters; real values give the formulas as written, and $x \rightarrow ix$, $k \rightarrow ik$ give the unitary version.

THREE ORDERINGS OF THE SAME EXPONENTIAL

Since $[xP, kQ] = xk[P, Q] = -i\hbar xk$ is a c-number, the Baker–Campbell–Hausdorff formula $e^A e^B = e^{A+B+\frac{1}{2}[A,B]}$ is exact:

$$e^{xP} e^{kQ} = e^{xP+kQ} e^{-i\hbar xk/2}, \quad e^{kQ} e^{xP} = e^{xP+kQ} e^{+i\hbar xk/2}.$$

The phases $e^{\mp i\hbar xk/2}$ are the ordering dependence. The symmetric splitting has none: apply BCH twice,

$$e^{\frac{kQ}{2}} e^{xP} e^{\frac{kQ}{2}} = e^{\frac{kQ}{2}} e^{xP+\frac{kQ}{2}} e^{-i\hbar xk/4} = e^{xP+kQ} e^{+i\hbar xk/4} e^{-i\hbar xk/4} = e^{xP+kQ}.$$

This is an identity, not an approximation: the Weyl-ordered exponential is the symmetric product. The same holds with P split instead, $e^{xP/2} e^{kQ} e^{xP/2} = e^{xP+kQ}$.

THE MIDPOINT RULE

Take matrix elements between position eigenstates, with $\langle q|p\rangle = e^{ipq/\hbar}/\sqrt{2\pi\hbar}$. The outer factors act on the bra and ket and give $e^{kq'/2}$ and $e^{kq/2}$:

$$\langle q'|e^{xP+kQ}|q\rangle = e^{k(q+q')/2} \langle q'|e^{xP}|q\rangle = \int \frac{dp}{2\pi\hbar} e^{ip(q'-q)/\hbar} e^{xp+k\bar{q}}, \quad \bar{q} = \frac{q+q'}{2}.$$

The classical function e^{xp+kq} appears evaluated at the *midpoint* $\bar{q}$. Because every Weyl-ordered operator is a superposition of such exponentials, the result holds for all of them:

$$\langle q'|f_W(Q, P)|q\rangle = \int \frac{dp}{2\pi\hbar} e^{ip(q'-q)/\hbar} f(\bar{q}, p).$$

Check with $f = qp$: $\langle q'|\frac{1}{2}(QP + PQ)|q\rangle = \frac{1}{2}(q' + q)\langle q'|P|q\rangle$, which is the midpoint value times the matrix element of P .

CONNECTION TO THE PATH INTEGRAL

Slice the time evolution into steps Δt and insert position and momentum eigenstates. If the Hamiltonian operator is written in Weyl-ordered form with symbol $H_W(q, p)$, each short-time matrix element is given by the midpoint rule, and the phase-space path integral reads

$$\int \mathcal{D}q \mathcal{D}p \exp \left\{ \frac{i}{\hbar} \int dt [p\dot{q} - H_W(q, p)] \right\},$$

with H_W evaluated at $\bar{q}_j = \frac{1}{2}(q_j + q_{j+1})$ on each slice.

- **The midpoint rule selects an ordering; it does not remove the ambiguity.** A discretization of the path integral corresponds to an ordering of the operator H . Different choices give Hamiltonians that differ by terms of order $\hbar$, and the midpoint rule is the one that corresponds to Weyl ordering.
- **For $H = \frac{P^2}{2m} + V(Q)$ there is nothing to choose.** No term mixes Q and P , H_W is the classical Hamiltonian, and evaluating V at q_j , q_{j+1} or $\bar{q}_j$ gives the same continuum limit.
- **When Q and P mix, the choice matters.** For a charged particle, $H = (P - eA(Q))^2/2m$ contains $PA + AP$, which is Weyl ordered; the $\dot{q} \cdot A$ term must be evaluated at the midpoint to obtain the gauge-invariant Schrödinger equation.

WEYL ORDERING IS NOT THE TROTTER SPLITTING

A different splitting is used to build the path integral in the first place. For $H = K + V$ with $K = P^2/2m$,

$$e^{-iH\Delta t/\hbar} = e^{-iK\Delta t/2\hbar} e^{-iV\Delta t/\hbar} e^{-iK\Delta t/2\hbar} + \mathcal{O}(\Delta t^3),$$

the symmetric Trotter (Strang) splitting. It is an approximation, exact only as $\Delta t \rightarrow 0$, because $[K, V]$ is an operator and higher BCH terms do not vanish; the errors, $\mathcal{O}(\Delta t^3)$ per step, sum to zero over the $T/\Delta t$ steps. The identity $e^{kQ/2} e^{xP} e^{kQ/2} = e^{xP+kQ}$ is different: it is exact, because $[Q, P]$ is a c-number.

SUMMARY

- Weyl ordering maps e^{xp+kq} to e^{xP+kQ} , and qp to $\frac{1}{2}(QP + PQ)$.
- $e^{kQ/2} e^{xP} e^{kQ/2} = e^{xP+kQ}$ exactly; the non-symmetric orderings differ from it by the phases $e^{\mp i\hbar xk/2}$.
- Matrix elements of Weyl-ordered operators put the classical function at the midpoint $\bar{q}$. In the path integral, the midpoint rule is the discretization that corresponds to Weyl ordering; for $P^2/2m + V(Q)$ any evaluation point gives the same continuum limit.